

```

% --- Task 1: Thermodynamic Potential of Methane/LOX Combustion ---
% This script calculates the standard enthalpy, entropy, and Gibbs Free Energy
% change for the complete combustion of methane with oxygen.

clear; clc; close all;

%% 1. Define Standard Thermodynamic Data (at 298.15 K and 1 bar)
% Source: NIST Chemistry WebBook. It is good practice to verify these values.
% All enthalpy values are in kJ/mol.
% All entropy values are in J/mol·K.

% --- Reactants ---
% CH4 (Methane, gas)
H_f_CH4 = -74.87; % Standard enthalpy of formation
S_CH4 = 186.25; % Standard molar entropy

% O2 (Oxygen, gas)
H_f_O2 = 0; % By definition for a pure element in its standard state
S_O2 = 205.15;

% --- Products ---
% CO2 (Carbon Dioxide, gas)
H_f_CO2 = -393.52;
S_CO2 = 213.79;

% H2O (Water, gas)
H_f_H2O = -241.83;
S_H2O = 188.84;

%% 2. Define the Stoichiometry of the Reaction
% Reaction: 1*CH4 + 2*O2 -> 1*CO2 + 2*H2O
n_CH4 = 1;
n_O2 = 2;
n_CO2 = 1;
n_H2O = 2;

%% 3. Perform the Thermodynamic Calculations

% --- Enthalpy Change (ΔH_rxn) ---
% ΔH_rxn = sum(H_products) - sum(H_reactants)
H_products = (n_CO2 * H_f_CO2) + (n_H2O * H_f_H2O)

H_products =
-877.1800

H_reactants = (n_CH4 * H_f_CH4) + (n_O2 * H_f_O2)

H_reactants =
-74.8700

delta_H_rxn = H_products - H_reactants % Result in kJ/mol

```

```
delta_H_rxn =  
-802.3100
```

```
% --- Entropy Change ( $\Delta S_{\text{rxn}}$ ) ---  
%  $\Delta S_{\text{rxn}} = \text{sum}(S_{\text{products}}) - \text{sum}(S_{\text{reactants}})$   
S_products = (n_CO2 * S_CO2) + (n_H2O * S_H2O)
```

```
S_products =  
591.4700
```

```
S_reactants = (n_CH4 * S_CH4) + (n_O2 * S_O2)
```

```
S_reactants =  
596.5500
```

```
delta_S_rxn = S_products - S_reactants % Result in J/mol·K
```

```
delta_S_rxn =  
-5.0800
```

```
% --- Gibbs Free Energy Change ( $\Delta G_{\text{rxn}}$ ) ---  
%  $\Delta G = \Delta H - T\Delta S$   
T_std = 298.15; % Standard temperature in Kelvin  
% Convert delta_S from J to kJ by dividing by 1000  
delta_G_rxn = delta_H_rxn - (T_std * (delta_S_rxn / 1000)) % Result in kJ/mol
```

```
delta_G_rxn =  
-800.7954
```

```
%% 4. Display the Results
```

```
fprintf('--- Thermodynamic Analysis of Methane Combustion ---\n');
```

```
--- Thermodynamic Analysis of Methane Combustion ---
```

```
fprintf('Reaction: %d CH4 + %d O2 -> %d CO2 + %d H2O\n', n_CH4, n_O2, n_CO2, n_H2O);
```

```
Reaction: 1 CH4 + 2 O2 -> 1 CO2 + 2 H2O
```

```
fprintf('Standard Enthalpy Change ( $\Delta H$ ): %.2f kJ/mol\n', delta_H_rxn);
```

```
Standard Enthalpy Change ( $\Delta H$ ): -802.31 kJ/mol
```

```
fprintf('Standard Entropy Change ( $\Delta S$ ): %.2f J/mol·K\n', delta_S_rxn);
```

```
Standard Entropy Change ( $\Delta S$ ): -5.08 J/mol·K
```

```
fprintf('Gibbs Free Energy Change ( $\Delta G$ ): %.2f kJ/mol\n', delta_G_rxn);
```

```
Gibbs Free Energy Change ( $\Delta G$ ): -800.80 kJ/mol
```

```
if delta_G_rxn < 0  
    fprintf('Result: The Gibbs Free Energy is highly negative, confirming the  
    reaction is very spontaneous.\n');
```

```

else
    fprintf('Result: The Gibbs Free Energy is positive, confirming the reaction is
not spontaneous.\n');
end

```

Result: The Gibbs Free Energy is highly negative, confirming the reaction is very spontaneous.

```

% --- Task 2: Visualizing the Activation Energy Barrier ---
% This script creates a conceptual plot showing the energy "hill" that
% uncatalyzed cryogenic propellants must overcome to react.

clear; clc; close all;

%% 1. Define the Reaction's Progress and Energy Levels
% We create a simple x-axis to show the reaction progressing from start to finish.
reaction_coordinate = linspace(0, 10, 500);

% Define the starting and ending energy levels based on our previous calculation.
energy_reactants = 0;      % Let's set the reactants as our baseline energy level
(0)
energy_products = -801;    % The energy released, from our Task 1 result (in kJ/mol)

%% 2. Model the "Energy Hill" (Activation Energy)
% This is a conceptual hill. We model it to be high, representing the
% large energy input required to break the strong O=O and C-H bonds.
Ea_height = 500;           % A representative high activation energy (in kJ/mol)
hill_position = 4;         % Where the peak of the hill is on our x-axis
hill_width = 1.5;         % How wide the hill is

% This equation calculates the energy profile: a line from reactants to
% products, with the activation energy hill added on top.
energy_profile = linspace(energy_reactants, energy_products, 500) ...
    + Ea_height * exp(-(reaction_coordinate - hill_position).^2 / (2 *
hill_width^2))

```

```

energy_profile = 1x500
    14.2828    13.1942    12.1217    11.0655    10.0260     9.0036     7.9987     7.0117 ...

```

```

%% 3. Visualize the Result
figure('Name', 'The Activation Energy Problem', 'Position', [100, 100, 900, 600]);
plot(reaction_coordinate, energy_profile, 'r-', 'LineWidth', 3);

hold on; % Hold the plot to add more elements

% Add an arrow and text to clearly label the Activation Energy
arrow_start = [hill_position, energy_reactants];
arrow_end = [hill_position, energy_reactants + Ea_height];
quiver(arrow_start(1), arrow_start(2), 0, arrow_end(2)-arrow_start(2), 0, ...
    'Color', 'k', 'LineWidth', 2, 'MaxHeadSize', 0.4);

```

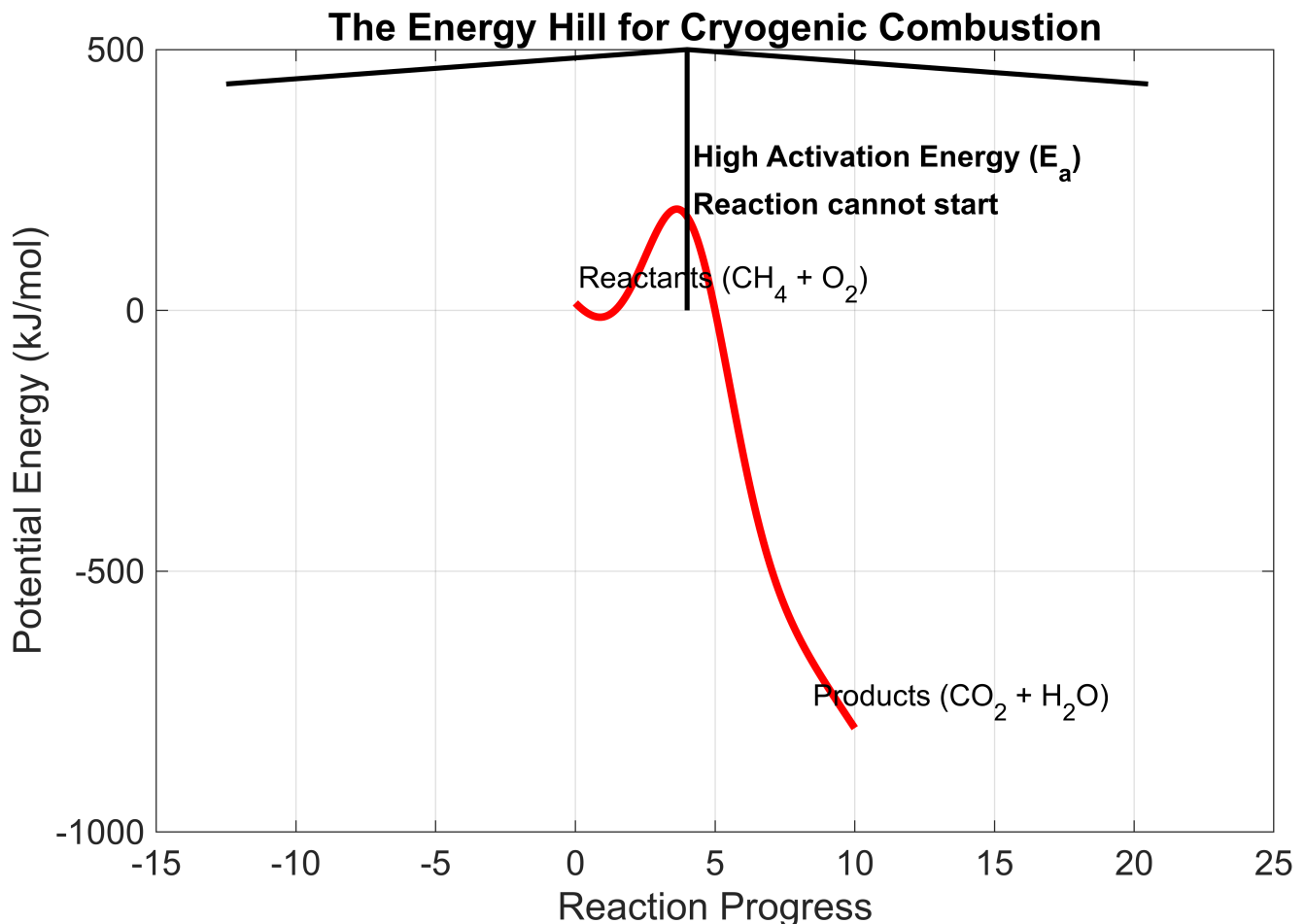
```

% Add labels
text(0.1, energy_reactants + 50, 'Reactants (CH4 + O2)', 'FontSize', 12);
text(8.5, energy_products + 50, 'Products (CO2 + H2O)', 'FontSize', 12);
text(hill_position + 0.2, energy_reactants + Ea_height/2, ...
    ['High Activation Energy (Ea)' char(10) 'Reaction cannot start'], 'FontSize',
    12, 'FontWeight', 'bold');

% Formatting
title('The Energy Hill for Cryogenic Combustion');
xlabel('Reaction Progress');
ylabel('Potential Energy (kJ/mol)');
grid on;
set(gca, 'FontSize', 14);
hold off;

% Export the high-quality figure for your report
exportgraphics(gcf, 'Activation_Energy_Problem.pdf', 'Resolution', 300);

```



```

% --- Task 3 (Revised): Calculation-Based Analysis of the Energy Barrier ---
% This script uses real bond energy values to compare the "upfront energy
% cost" of a thermal ignition path versus a plasma-assisted path.

```

```
clear; clc; close all;

%% Step 1: Define Key Chemical Bond Energies (The Data)
% Source: Standard Chemistry Data (NIST, CRC Handbook)
O_O_double_bond = 498; % Energy to break O=O in O2 (kJ/mol)
C_H_single_bond = 413; % Average energy to break a C-H bond in CH4 (kJ/mol)

% Activation energy for a key radical reaction
% Source: Combustion chemistry literature (e.g., for O• + CH4)
Ea_radical_reaction = 45; % Activation energy for the next step (kJ/mol)

fprintf('--- TASK 3: A Calculation-Based Analysis of the Energy Barrier ---\n');
```

--- TASK 3: A Calculation-Based Analysis of the Energy Barrier ---

```
%% Step 2: SCENARIO A - The Thermal Ignition Path
fprintf('** SCENARIO A: The Thermal Ignition Path **\n\n');
```

** SCENARIO A: The Thermal Ignition Path **

```
fprintf('Assumption: In a purely thermal model, the reaction must overcome the
energy\n');
```

Assumption: In a purely thermal model, the reaction must overcome the energy

```
fprintf('cost of breaking ALL initial bonds to begin.\n\n');
```

cost of breaking ALL initial bonds to begin.

```
% The overall reaction requires breaking bonds in 1x CH4 and 2x O2
num_CH_bonds = 4;
num_OO_bonds = 2;

% CALCULATION: Total energy required to break all reactant bonds
total_thermal_cost = (num_CH_bonds * C_H_single_bond) + (num_OO_bonds *
O_O_double_bond);

fprintf('    Energy to break 4 C-H bonds: 4 * %.0f = %.0f kJ/mol\n',
C_H_single_bond, num_CH_bonds * C_H_single_bond);
```

Energy to break 4 C-H bonds: 4 * 413 = 1652 kJ/mol

```
fprintf('    Energy to break 2 O=O bonds: 2 * %.0f = %.0f kJ/mol\n',
O_O_double_bond, num_OO_bonds * O_O_double_bond);
```

Energy to break 2 O=O bonds: 2 * 498 = 996 kJ/mol

```
fprintf('    TOTAL UPFRONT ENERGY COST (Thermal Path): %.0f kJ/mol\n\n',
total_thermal_cost);
```

TOTAL UPFRONT ENERGY COST (Thermal Path): 2648 kJ/mol

```
fprintf('STATEMENT: This massive upfront energy cost represents the very high\n');
```

STATEMENT: This massive upfront energy cost represents the very high

```
fprintf('          activation barrier for the thermal reaction.\n');
```

activation barrier for the thermal reaction.

```
%% Step 3: SCENARIO B - The Plasma-Assisted Path
```

```
fprintf('** SCENARIO B: The Plasma-Assisted Path **\n\n');
```

```
** SCENARIO B: The Plasma-Assisted Path **
```

```
fprintf('Assumption: Plasma uses its electrical energy to perform a targeted  
attack,\n');
```

Assumption: Plasma uses its electrical energy to perform a targeted attack,

```
fprintf('breaking the strongest bonds first to create reactive radicals.\n\n');
```

breaking the strongest bonds first to create reactive radicals.

```
fprintf('    Plasma''s Primary Target: The 2 O=O double bonds.\n');
```

Plasma's Primary Target: The 2 O=O double bonds.

```
fprintf('    CALCULATION: Energy cost for this targeted step = %.0f kJ/mol.\n\n',  
num_OO_bonds * O_O_double_bond);
```

CALCULATION: Energy cost for this targeted step = 996 kJ/mol.

```
fprintf('This creates atomic oxygen radicals (O•). The NEXT reaction step is  
now:\n');
```

This creates atomic oxygen radicals (O•). The NEXT reaction step is now:

```
fprintf('    O• + CH4  -->  •OH + •CH3\n\n');
```

O• + CH4 --> •OH + •CH3

```
fprintf('    The activation energy for THIS subsequent step is known to be very  
low.\n');
```

The activation energy for THIS subsequent step is known to be very low.

```
fprintf('    ACTIVATION ENERGY for Radical Step: %.0f kJ/mol\n\n',  
Ea_radical_reaction);
```

ACTIVATION ENERGY for Radical Step: 45 kJ/mol

```
fprintf('STATEMENT: The plasma bypasses the massive %.0f kJ/mol thermal barrier.\n', total_thermal_cost);
```

STATEMENT: The plasma bypasses the massive 2648 kJ/mol thermal barrier.

```
fprintf('                Instead, it creates a new pathway where the highest\n');
```

Instead, it creates a new pathway where the highest

```
fprintf('                energy hurdle is only %.0f kJ/mol.\n', Ea_radical_reaction);
```

energy hurdle is only 45 kJ/mol.

```
% --- Task 4: Visualizing the Plasma-Catalyzed Solution ---
```

```
% This script updates the previous plot to show both the uncatalyzed path  
% and the new, lower-energy catalyzed path created by plasma.
```

```
clear; clc; close all;
```

```
%% 1. Define Reaction Progress and Energy Levels (Same as Task 2)
```

```
reaction_coordinate = linspace(0, 10, 500);
```

```
energy_reactants = 0;
```

```
energy_products = -801;
```

```
%% 2. Model the Uncatalyzed Path (The "Problem")
```

```
Ea_uncatalyzed = 500;
```

```
hill_position_uncat = 4;
```

```
hill_width_uncat = 1.5;
```

```
energy_profile_uncat = linspace(energy_reactants, energy_products, 500) ...  
    + Ea_uncatalyzed * exp(-(reaction_coordinate - hill_position_uncat).^2 / (2 *  
    hill_width_uncat^2))
```

```
energy_profile_uncat = 1x500
```

```
    14.2828    13.1942    12.1217    11.0655    10.0260     9.0036     7.9987     7.0117 ...
```

```
%% 3. Model the Plasma-Catalyzed Path (The "Solution")
```

```
% We model this with two smaller, sequential humps representing the  
% easy reactions between radicals.
```

```
Ea_cat1_height = 80;           % Height of the first small hill
```

```
hill_pos_cat1 = 2.5;
```

```
hill_width_cat1 = 0.8;
```

```
Ea_cat2_height = 60;           % Height of the second small hill
```

```
hill_pos_cat2 = 6.5;
```

```
hill_width_cat2 = 1.0;
```

```
% Calculate the energy profile for the new path
```

```
energy_profile_cat = linspace(energy_reactants, energy_products, 500) ...  
    + Ea_cat1_height * exp(-(reaction_coordinate - hill_pos_cat1).^2 / (2 *  
    hill_width_cat1^2)) ...
```

```

+ Ea_cat2_height * exp(-(reaction_coordinate - hill_pos_cat2).^2 / (2 *
hill_width_cat2^2))

```

```

energy_profile_cat = 1x500
    0.6061    -0.9500    -2.5025    -4.0513    -5.5961    -7.1367    -8.6728    -10.2042 ...

```

```

%% 4. Visualize Both Paths for a Clear Comparison

```

```

figure('Name', 'Problem vs. Solution', 'Position', [100, 100, 900, 600]);
hold on;

```

```

% Plot the uncatalyzed path (the problem)

```

```

plot(reaction_coordinate, energy_profile_uncat, 'r--', 'LineWidth', 3,
'DisplayName', 'Uncatalyzed Path (High E_a)');

```

```

% Plot the NEW catalyzed path (the solution)

```

```

plot(reaction_coordinate, energy_profile_cat, 'b-', 'LineWidth', 3, 'DisplayName',
'Plasma-Catalyzed Path (Low E_a)');

```

```

% Add labels

```

```

text(0.1, energy_reactants + 50, 'Reactants', 'FontSize', 12);
text(8.5, energy_products + 50, 'Products', 'FontSize', 12);

```

```

% Formatting

```

```

title('Plasma Creates a New, Lower-Energy Reaction Pathway');
xlabel('Reaction Progress');
ylabel('Potential Energy (kJ/mol)');
legend('show', 'Location', 'northeast');
grid on;
set(gca, 'FontSize', 14);
hold off;

```

```

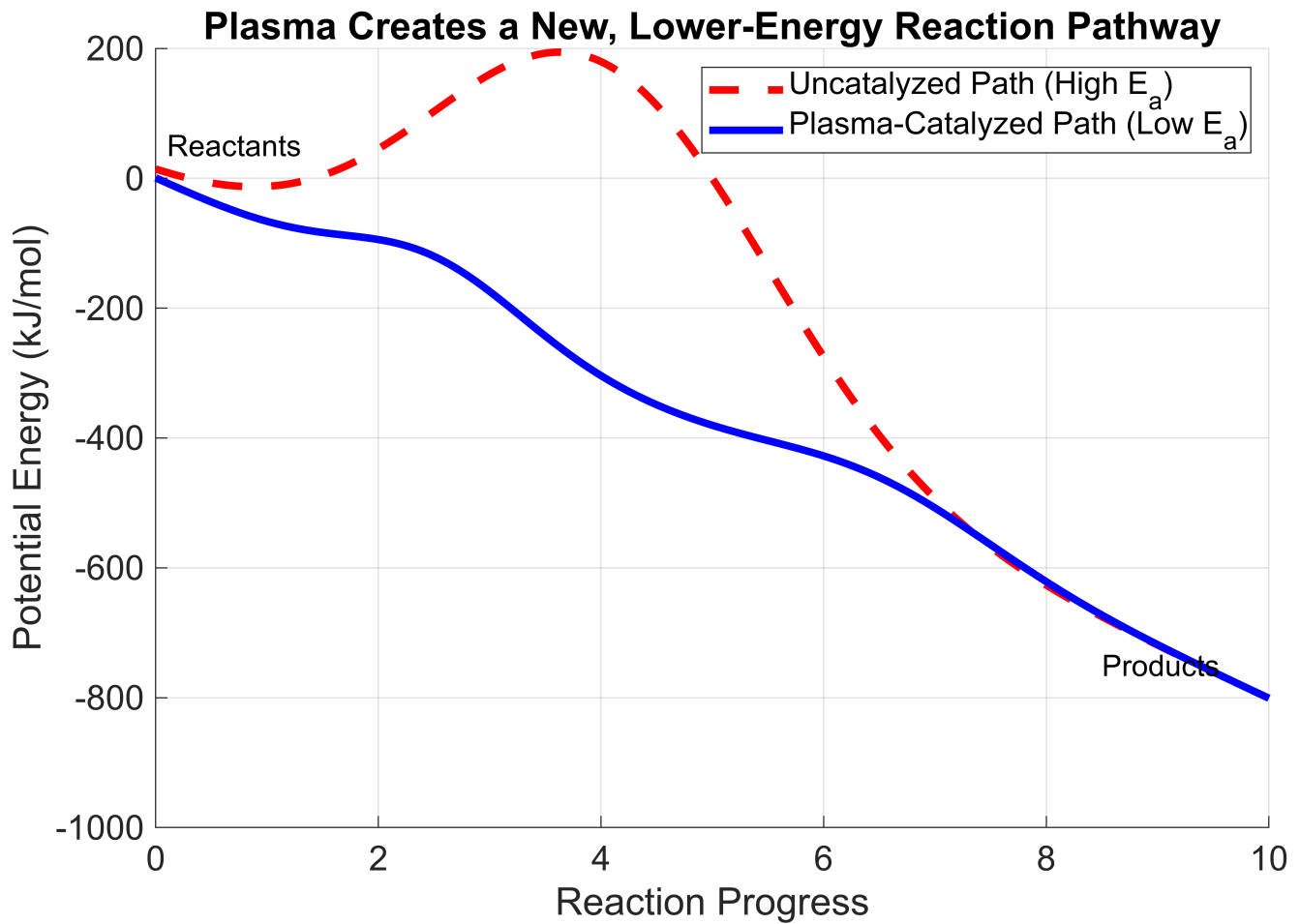
% Export the high-quality figure for your report

```

```

exportgraphics(gcf, 'Activation_Energy_Solution.pdf', 'Resolution', 300);

```



```
% --- Task 5 (Revised): A Detailed Analysis of Engine Performance (Isp) ---
% This script provides a comprehensive calculation and explanation of Specific
% Impulse (Isp) for a CH4/LOX engine, detailing the advantages and
% limitations of operating at different mixture ratios.
```

```
clear; clc; close all;
```

```
%% Step 1: The Core Concept of Specific Impulse (Isp)
fprintf('--- TASK 5: A DETAILED ANALYSIS OF ENGINE PERFORMANCE (Isp) ---\n');
```

```
--- TASK 5: A DETAILED ANALYSIS OF ENGINE PERFORMANCE (Isp) ---
```

```
fprintf('This lesson explains the most important metric for rocket engine
efficiency.\n');
```

```
This lesson explains the most important metric for rocket engine efficiency.
```

```
fprintf('** What is Specific Impulse (Isp)? **\n\n');
```

```
** What is Specific Impulse (Isp)? **
```

```
fprintf('Think of Isp as the engine''s "fuel efficiency" or "miles per gallon" (MPG).\n');
```

Think of Isp as the engine's "fuel efficiency" or "miles per gallon" (MPG).

```
fprintf('A higher Isp means the engine produces more thrust for the same amount of fuel.\n');
```

A higher Isp means the engine produces more thrust for the same amount of fuel.

```
fprintf('It tells us how efficiently the engine uses propellant.\n\n');
```

It tells us how efficiently the engine uses propellant.

```
fprintf('** The Physics Behind Isp **\n\n');
```

** The Physics Behind Isp **

```
fprintf('A simplified version of the ideal Isp equation is:\n\n');
```

A simplified version of the ideal Isp equation is:

```
fprintf('    Isp is proportional to sqrt( T_c / M_exhaust )\n\n');
```

Isp is proportional to $\sqrt{T_c / M_{\text{exhaust}}}$

```
fprintf('This reveals the most important trade-off in rocket engine design:\n');
```

This reveals the most important trade-off in rocket engine design:

```
fprintf('To get a high Isp, you want two things:\n');
```

To get a high Isp, you want two things:

```
fprintf('    1. A VERY HIGH combustion temperature (T_c).\n');
```

1. A VERY HIGH combustion temperature (T_c).

```
fprintf('    2. A VERY LOW average molar mass of your exhaust gas (M_exhaust).\n\n');
```

2. A VERY LOW average molar mass of your exhaust gas (M_{exhaust}).

```
fprintf('Our simulation will show how changing the fuel/oxidizer ratio affects these values.\n');
```

Our simulation will show how changing the fuel/oxidizer ratio affects these values.

```
%% Step 2: Define Realistic Constants and Assumptions
```

```
g0 = 9.81; % Standard gravity (m/s^2)
```

```
R_universal = 8314; % Universal gas constant (J/kmol.K)
```

```
T_c = 3600; % Realistic combustion chamber temperature (K) for CH4/O2
```

```
% Molar Masses (kg/kmol)
```

```
M_CH4 = 16.04; % Methane (Fuel)
```

```

M_O2 = 32.00;           % Oxygen (Oxidizer)
M_CO2 = 44.01;          % Carbon Dioxide
M_H2O = 18.02;          % Water
M_CO = 28.01;           % Carbon Monoxide
M_H2 = 2.02;            % Hydrogen gas (very light!)

% Assumed Specific Heat Ratio (gamma) of exhaust gases
gamma = 1.22;

%% Step 3: Calculation Across a Range of Mixture Ratios
fprintf('Now, we will run the calculation for different Mixture Ratios (MR).\n');

```

Now, we will run the calculation for different Mixture Ratios (MR).

```

fprintf('MR = mass of oxidizer / mass of fuel.\n\n');

```

MR = mass of oxidizer / mass of fuel.

```

% The stoichiometric (chemically perfect) MR for CH4 + 2O2 is 4.0.

```

```

mr_range = 2.5:0.05:4.5;
isp_values = zeros(size(mr_range))

```

```

isp_values = 1x41
    0    0    0    0    0    0    0    0    0    0    0    0 ...

```

```

M_exhaust_values = zeros(size(mr_range))

```

```

M_exhaust_values = 1x41
    0    0    0    0    0    0    0    0    0    0    0    0 ...

```

```

for i = 1:length(mr_range)
    mr = mr_range(i);
    moles_CH4 = 1;
    moles_O2 = mr * moles_CH4 * (M_CH4 / M_O2)

    if moles_O2 >= 2.0 % --- Stoichiometric or Fuel-Lean (MR >= 4.0) ---
        % In this case, we have excess oxygen.
        % Products: CO2, H2O, and heavy, unreacted O2.
        moles_CO2_prod = 1;
        moles_H2O_prod = 2;
        moles_O2_excess = moles_O2 - 2;
        total_mass_prod = moles_CO2_prod*M_CO2 + moles_H2O_prod*M_H2O +
        moles_O2_excess*M_O2
        total_moles_prod = moles_CO2_prod + moles_H2O_prod + moles_O2_excess
    else % --- Fuel-Rich (MR < 4.0) ---
        % Here, we don't have enough oxygen to burn all the fuel.
        % The combustion is incomplete, producing lighter molecules like CO and H2.
        moles_CO2_prod = moles_O2 - 1;
        moles_H2O_prod = 2;
        moles_CO_prod = 2 - moles_O2;
        moles_H2_prod = 2 - moles_O2;
    end
end

```

```

        total_mass_prod = moles_CO2_prod*M_CO2 + moles_H2O_prod*M_H2O +
        moles_CO_prod*M_CO + moles_H2_prod*M_H2
        total_moles_prod = moles_CO2_prod + moles_H2O_prod + moles_CO_prod +
        moles_H2_prod
    end

    % Calculate the average molar mass of the exhaust gas mixture
    M_exhaust = total_mass_prod / total_moles_prod
    M_exhaust_values(i) = M_exhaust

    % Calculate Isp using the Ideal Rocket Equation
    exhaust_velocity = sqrt((2 * gamma / (gamma - 1)) * (R_universal / M_exhaust) *
    T_c)
    isp_values(i) = exhaust_velocity / g0
end

```

```

moles_O2 =
1.2531
total_mass_prod =
69.6087
total_moles_prod =
3.7469
M_exhaust =
18.5778
M_exhaust_values = 1x41
    18.5778     0         0         0         0         0         0         0 ...
exhaust_velocity =
4.2271e+03
isp_values = 1x41
    430.8972     0         0         0         0         0         0         0 ...
moles_O2 =
1.2782
total_mass_prod =
69.9591
total_moles_prod =
3.7218
M_exhaust =
18.7970
M_exhaust_values = 1x41
    18.5778    18.7970         0         0         0         0         0         0 ...
exhaust_velocity =
4.2024e+03
isp_values = 1x41
    430.8972    428.3769         0         0         0         0         0         0 ...
moles_O2 =
1.3033
total_mass_prod =
70.3094
total_moles_prod =
3.6967
M_exhaust =
19.0193
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193         0         0         0         0         0 ...
exhaust_velocity =
4.1778e+03
isp_values = 1x41
    430.8972    428.3769    425.8671         0         0         0         0         0 ...
moles_O2 =
1.3283

```

```

total_mass_prod =
70.6598
total_moles_prod =
3.6717
M_exhaust =
19.2445
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445         0         0         0         0 ...
exhaust_velocity =
4.1532e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674         0         0         0         0 ...
moles_O2 =
1.3534
total_mass_prod =
71.0102
total_moles_prod =
3.6466
M_exhaust =
19.4729
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729         0         0         0 ...
exhaust_velocity =
4.1288e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778         0         0         0 ...
moles_O2 =
1.3784
total_mass_prod =
71.3606
total_moles_prod =
3.6216
M_exhaust =
19.7044
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044         0         0 ...
exhaust_velocity =
4.1045e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981         0         0 ...
moles_O2 =
1.4035
total_mass_prod =
71.7109
total_moles_prod =
3.5965
M_exhaust =
19.9391
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391         0 ...
exhaust_velocity =
4.0803e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280         0 ...
moles_O2 =
1.4286
total_mass_prod =
72.0613
total_moles_prod =
3.5714
M_exhaust =
20.1771
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...

```

```

exhaust_velocity =
4.0561e+03
isp_values = 1x41
  430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
1.4536
total_mass_prod =
72.4117
total_moles_prod =
3.5464
M_exhaust =
20.4185
M_exhaust_values = 1x41
  18.5778  18.7970  19.0193  19.2445  19.4729  19.7044  19.9391  20.1771 ...
exhaust_velocity =
4.0321e+03
isp_values = 1x41
  430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
1.4787
total_mass_prod =
72.7621
total_moles_prod =
3.5213
M_exhaust =
20.6633
M_exhaust_values = 1x41
  18.5778  18.7970  19.0193  19.2445  19.4729  19.7044  19.9391  20.1771 ...
exhaust_velocity =
4.0081e+03
isp_values = 1x41
  430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
1.5037
total_mass_prod =
73.1124
total_moles_prod =
3.4962
M_exhaust =
20.9117
M_exhaust_values = 1x41
  18.5778  18.7970  19.0193  19.2445  19.4729  19.7044  19.9391  20.1771 ...
exhaust_velocity =
3.9842e+03
isp_values = 1x41
  430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
1.5288
total_mass_prod =
73.4628
total_moles_prod =
3.4712
M_exhaust =
21.1636
M_exhaust_values = 1x41
  18.5778  18.7970  19.0193  19.2445  19.4729  19.7044  19.9391  20.1771 ...
exhaust_velocity =
3.9605e+03
isp_values = 1x41
  430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
1.5539
total_mass_prod =
73.8132

```

```

total_moles_prod =
3.4461
M_exhaust =
21.4192
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.9368e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.5789
total_mass_prod =
74.1635
total_moles_prod =
3.4211
M_exhaust =
21.6785
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.9131e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.6040
total_mass_prod =
74.5139
total_moles_prod =
3.3960
M_exhaust =
21.9417
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.8896e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.6291
total_mass_prod =
74.8643
total_moles_prod =
3.3709
M_exhaust =
22.2087
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.8661e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.6541
total_mass_prod =
75.2147
total_moles_prod =
3.3459
M_exhaust =
22.4798
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.8428e+03

```

```

isp_values = 1x41
  430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
1.6792
total_mass_prod =
75.5650
total_moles_prod =
3.3208
M_exhaust =
22.7550
M_exhaust_values = 1x41
  18.5778  18.7970  19.0193  19.2445  19.4729  19.7044  19.9391  20.1771 ...
exhaust_velocity =
3.8195e+03
isp_values = 1x41
  430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
1.7042
total_mass_prod =
75.9154
total_moles_prod =
3.2957
M_exhaust =
23.0343
M_exhaust_values = 1x41
  18.5778  18.7970  19.0193  19.2445  19.4729  19.7044  19.9391  20.1771 ...
exhaust_velocity =
3.7962e+03
isp_values = 1x41
  430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
1.7293
total_mass_prod =
76.2658
total_moles_prod =
3.2707
M_exhaust =
23.3180
M_exhaust_values = 1x41
  18.5778  18.7970  19.0193  19.2445  19.4729  19.7044  19.9391  20.1771 ...
exhaust_velocity =
3.7731e+03
isp_values = 1x41
  430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
1.7544
total_mass_prod =
76.6162
total_moles_prod =
3.2456
M_exhaust =
23.6060
M_exhaust_values = 1x41
  18.5778  18.7970  19.0193  19.2445  19.4729  19.7044  19.9391  20.1771 ...
exhaust_velocity =
3.7500e+03
isp_values = 1x41
  430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
1.7794
total_mass_prod =
76.9665
total_moles_prod =
3.2206

```

```

M_exhaust =
23.8985
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.7270e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.8045
total_mass_prod =
77.3169
total_moles_prod =
3.1955
M_exhaust =
24.1956
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.7040e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.8296
total_mass_prod =
77.6673
total_moles_prod =
3.1704
M_exhaust =
24.4973
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.6811e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.8546
total_mass_prod =
78.0177
total_moles_prod =
3.1454
M_exhaust =
24.8039
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.6583e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.8797
total_mass_prod =
78.3680
total_moles_prod =
3.1203
M_exhaust =
25.1154
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.6355e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...

```

```

moles_O2 =
1.9047
total_mass_prod =
78.7184
total_moles_prod =
3.0953
M_exhaust =
25.4320
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.6128e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.9298
total_mass_prod =
79.0688
total_moles_prod =
3.0702
M_exhaust =
25.7537
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5902e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.9549
total_mass_prod =
79.4192
total_moles_prod =
3.0451
M_exhaust =
26.0808
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5676e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
1.9799
total_mass_prod =
79.7695
total_moles_prod =
3.0201
M_exhaust =
26.4132
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5451e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
2.0050
total_mass_prod =
80.2100
total_moles_prod =
3.0050
M_exhaust =
26.6922

```

```

M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5265e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
2.0301
total_mass_prod =
81.0120
total_moles_prod =
3.0301
M_exhaust =
26.7361
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5236e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
2.0551
total_mass_prod =
81.8140
total_moles_prod =
3.0551
M_exhaust =
26.7793
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5208e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
2.0802
total_mass_prod =
82.6160
total_moles_prod =
3.0802
M_exhaust =
26.8217
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5180e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
2.1052
total_mass_prod =
83.4180
total_moles_prod =
3.1052
M_exhaust =
26.8635
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5153e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
2.1303

```

```

total_mass_prod =
84.2200
total_moles_prod =
3.1303
M_exhaust =
26.9047
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5126e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
2.1554
total_mass_prod =
85.0220
total_moles_prod =
3.1554
M_exhaust =
26.9451
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5099e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
2.1804
total_mass_prod =
85.8240
total_moles_prod =
3.1804
M_exhaust =
26.9850
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5073e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
2.2055
total_mass_prod =
86.6260
total_moles_prod =
3.2055
M_exhaust =
27.0242
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...
exhaust_velocity =
3.5048e+03
isp_values = 1x41
    430.8972    428.3769    425.8671    423.3674    420.8778    418.3981    415.9280    413.4674 ...
moles_O2 =
2.2306
total_mass_prod =
87.4280
total_moles_prod =
3.2306
M_exhaust =
27.0628
M_exhaust_values = 1x41
    18.5778    18.7970    19.0193    19.2445    19.4729    19.7044    19.9391    20.1771 ...

```

```

exhaust_velocity =
3.5023e+03
isp_values = 1x41
    430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...
moles_O2 =
2.2556
total_mass_prod =
88.2300
total_moles_prod =
3.2556
M_exhaust =
27.1008
M_exhaust_values = 1x41
    18.5778  18.7970  19.0193  19.2445  19.4729  19.7044  19.9391  20.1771 ...
exhaust_velocity =
3.4998e+03
isp_values = 1x41
    430.8972  428.3769  425.8671  423.3674  420.8778  418.3981  415.9280  413.4674 ...

```

```
fprintf('Calculation complete. Generating plot and analysis...\n');
```

Calculation complete. Generating plot and analysis...

```

% Step 4: Visualize the Results with High-Quality Formatting
figure('Name', 'Engine Performance Analysis', 'Position', [100, 100, 900, 600]);

plot(mr_range, isp_values, 'b-', 'LineWidth', 3, 'DisplayName', 'Calculated
I_{sp}');
hold on;

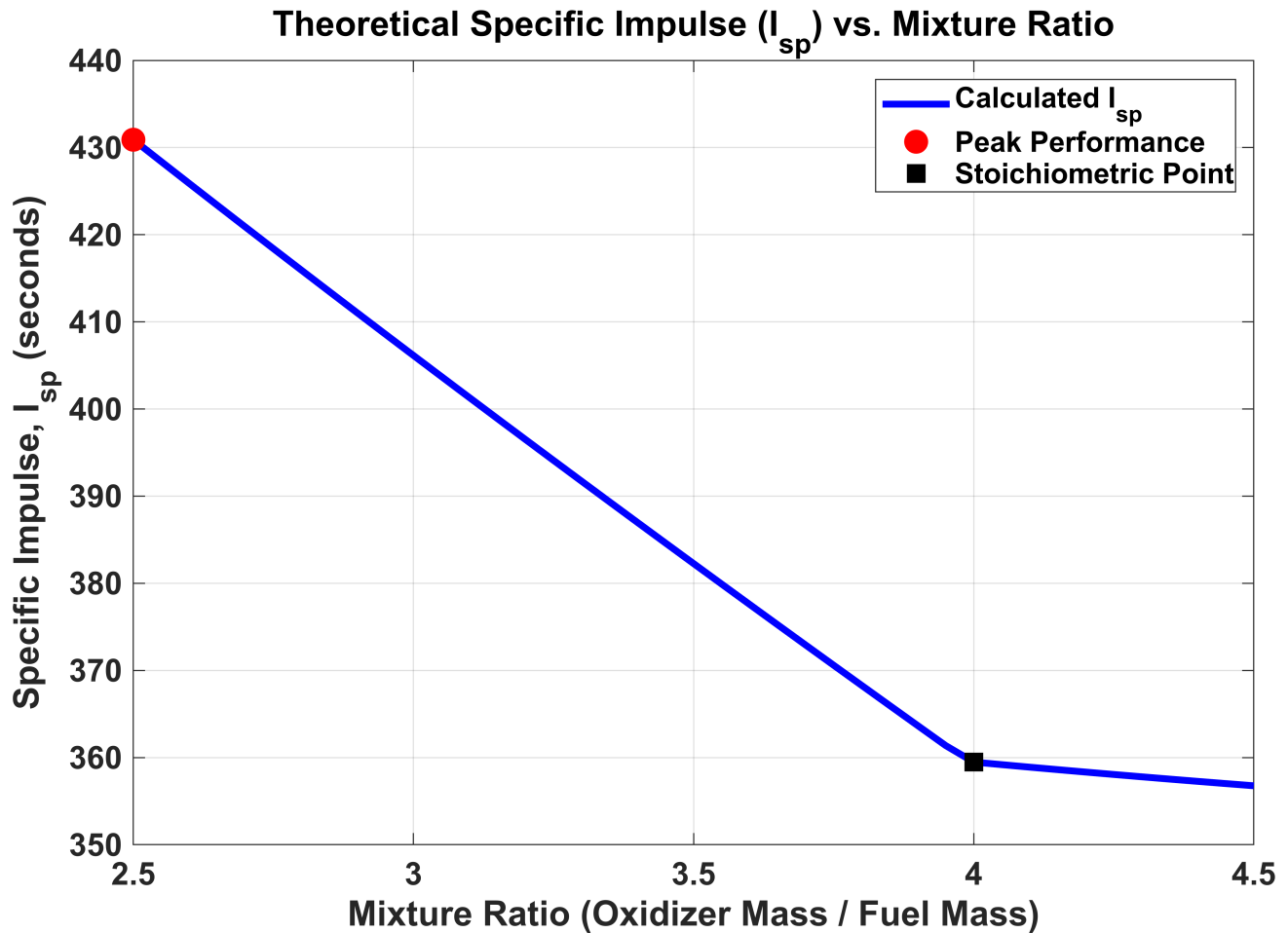
% Find and mark the peak Isp
[peak_isp, idx] = max(isp_values);
peak_mr = mr_range(idx);
plot(peak_mr, peak_isp, 'ro', 'MarkerSize', 10, 'MarkerFaceColor', 'r',
'DisplayName', 'Peak Performance');

% Mark the stoichiometric point for reference
stoich_mr = 4.0;
stoich_isp_idx = find(mr_range == stoich_mr);
plot(stoich_mr, isp_values(stoich_isp_idx), 'ks', 'MarkerSize', 10,
'MarkerFaceColor', 'k', 'DisplayName', 'Stoichiometric Point');

hold off;
title('Theoretical Specific Impulse (I_{sp}) vs. Mixture Ratio');
xlabel('Mixture Ratio (Oxidizer Mass / Fuel Mass)');
ylabel('Specific Impulse, I_{sp} (seconds)');
legend('show', 'Location', 'northeast');
grid on;
set(gca, 'FontSize', 14, 'FontWeight', 'bold');

exportgraphics(gcf, 'Engine_Isp_Analysis.pdf', 'Resolution', 300);

```



`% Step 5: Explanation and Analysis of the Graph`

```
fprintf('** ANALYSIS OF THE GRAPH: What This Result Means **\n\n');
```

```
** ANALYSIS OF THE GRAPH: What This Result Means **
```

```
fprintf('The plot clearly shows that the best performance is NOT at the\n');
```

The plot clearly shows that the best performance is NOT at the

```
fprintf('chemically perfect (stoichiometric) point of MR = 4.0.\n');
```

chemically perfect (stoichiometric) point of MR = 4.0.

```
fprintf('The peak Isp of %.1f s occurs at a FUEL-RICH mixture ratio of MR = %.2f.\n\n', peak_isp, peak_mr);
```

The peak Isp of 430.9 s occurs at a FUEL-RICH mixture ratio of MR = 2.50.

```
fprintf('** Advantage of Running Fuel-Rich (The "Why") **\n');
```

```
** Advantage of Running Fuel-Rich (The "Why") **
```

```
fprintf('    - HIGHER EFFICIENCY: When there is not enough oxygen, the  
combustion\n');
```

- HIGHER EFFICIENCY: When there is not enough oxygen, the combustion

```
fprintf('        produces very lightweight Hydrogen gas (H2, molar mass ~2) in the  
exhaust.\n');
```

produces very lightweight Hydrogen gas (H2, molar mass ~2) in the exhaust.

```
fprintf('        This drastically LOWERS the average molar mass (M_exhaust) of the  
exhaust gases.\n');
```

This drastically LOWERS the average molar mass (M_exhaust) of the exhaust gases.

```
fprintf('        According to our key equation, a lower M_exhaust leads to a HIGHER  
Isp.\n');
```

According to our key equation, a lower M_exhaust leads to a HIGHER Isp.

```
fprintf('    - COOLER COMBUSTION: Running fuel-rich leads to a slightly lower  
combustion\n');
```

- COOLER COMBUSTION: Running fuel-rich leads to a slightly lower combustion

```
fprintf('        temperature, which reduces the thermal stress on the engine nozzle  
and chamber.\n\n');
```

temperature, which reduces the thermal stress on the engine nozzle and chamber.

```
fprintf('** Limitations and Trade-offs **\n');
```

** Limitations and Trade-offs **

```
fprintf('    - LOWER ABSOLUTE THRUST: While more efficient, fuel-rich combustion  
releases\n');
```

- LOWER ABSOLUTE THRUST: While more efficient, fuel-rich combustion releases

```
fprintf('        slightly less total energy, which can result in a lower absolute  
thrust.\n');
```

slightly less total energy, which can result in a lower absolute thrust.

```
fprintf('    - INCOMPLETE COMBUSTION: You are essentially "wasting" a small amount  
of fuel,\n');
```

- INCOMPLETE COMBUSTION: You are essentially "wasting" a small amount of fuel,

```
fprintf('        which gets ejected as unburnt CO and H2.\n\n');
```

which gets ejected as unburnt CO and H2.

```
fprintf('CONCLUSION: This simulation proves why most real-world cryogenic engines  
are\n');
```

CONCLUSION: This simulation proves why most real-world cryogenic engines are

```
fprintf('designed to run slightly fuel-rich. They sacrifice a small amount of  
total\n');
```

designed to run slightly fuel-rich. They sacrifice a small amount of total

```
fprintf('thrust to gain a significant boost in fuel efficiency (Isp), allowing  
the\n');
```

thrust to gain a significant boost in fuel efficiency (Isp), allowing the

```
fprintf('rocket to achieve a higher final velocity.\n');
```

rocket to achieve a higher final velocity.